01_qc

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- QC Report- Nanopore Sequencing
  - Sample: T1
  - Sample: M1
  - Sample: A1
- Method:
  - QC pipeline steps:
- Analysis:

QC Report- Nanopore Sequencing

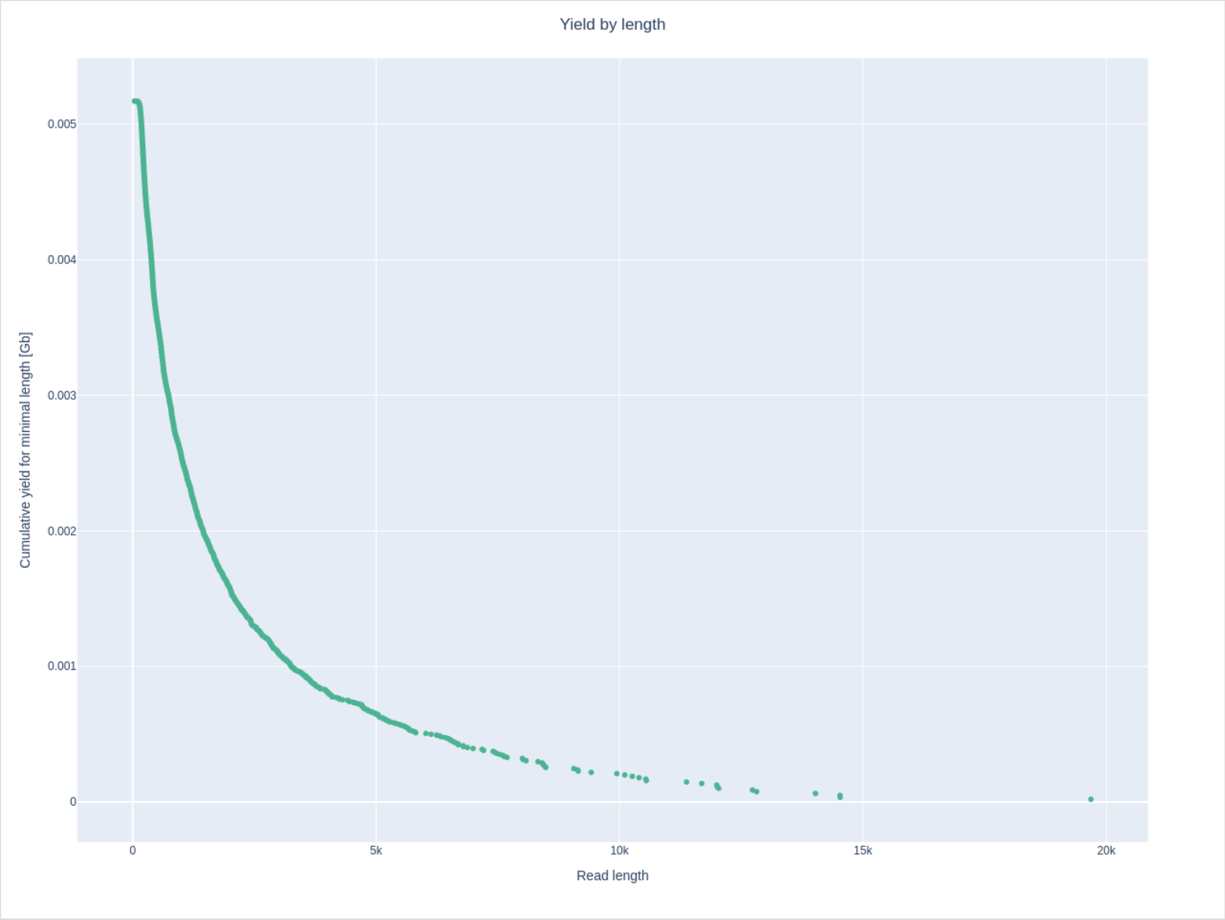
Sample: T1

Pre-Filtration QC

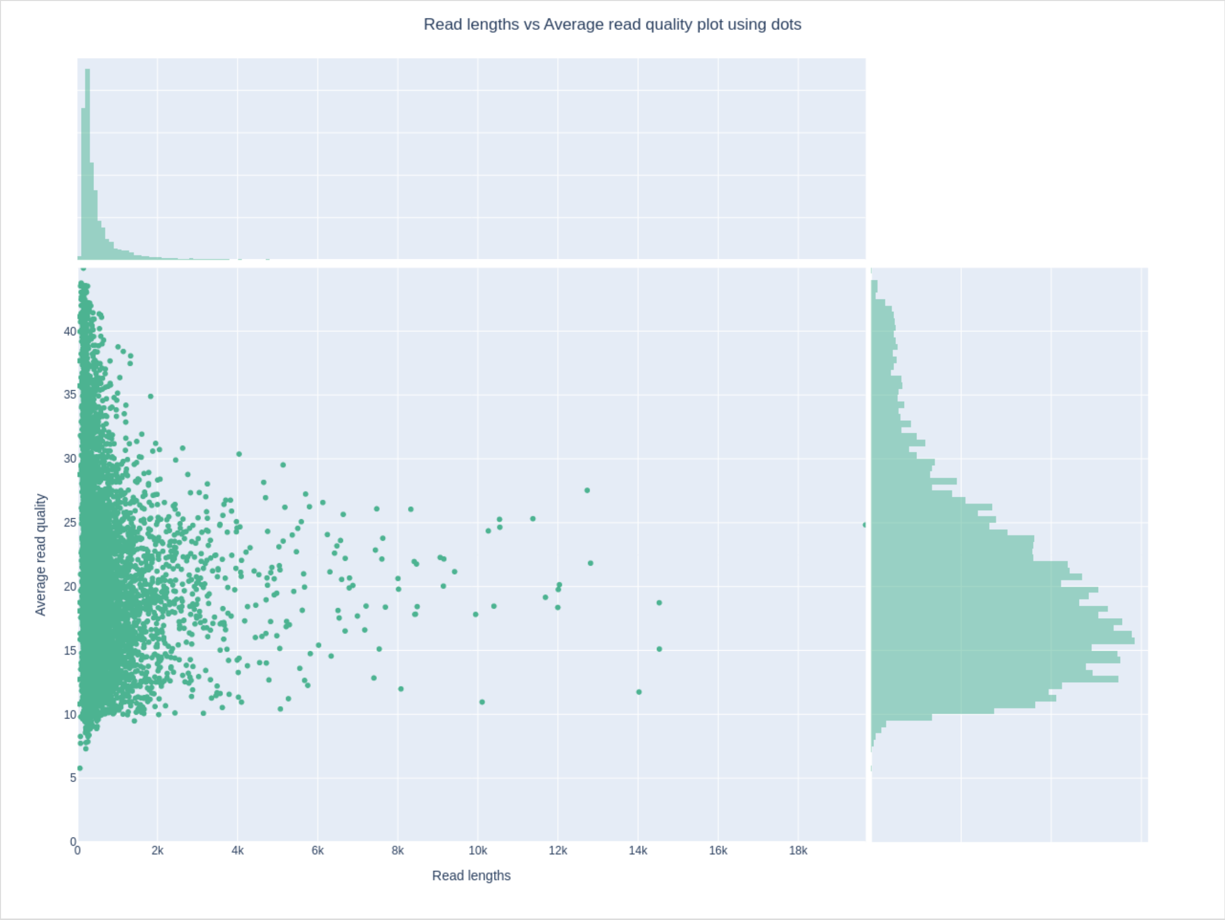
Statistics

```
General summary:
Mean read length:                 598.9
Mean read quality:                 16.3
Median read length:               321.0
Median read quality:               18.7
Number of reads:                8,634.0
Read length N50:                  978.0
STDEV read length:                965.8
Total bases:                5,170,693.0
Number, percentage and megabases of reads above quality cutoffs
>Q10:   8528 (98.8%) 5.1Mb
>Q15:   6287 (72.8%) 3.9Mb
>Q20:   3673 (42.5%) 2.3Mb
>Q25:   1788 (20.7%) 0.8Mb
>Q30:   834 (9.7%) 0.3Mb
Top 5 highest mean basecall quality scores and their read lengths
1:  45.0 (149)
2:  43.8 (93)
3:  43.7 (113)
4:  43.6 (134)
5:  43.6 (153)
Top 5 longest reads and their mean basecall quality score
1:  19684 (24.8)
2:  14534 (15.1)
3:  14530 (18.7)
4:  14026 (11.8)
5:  12816 (21.8)
```

Yield by Length



Length vs Quality

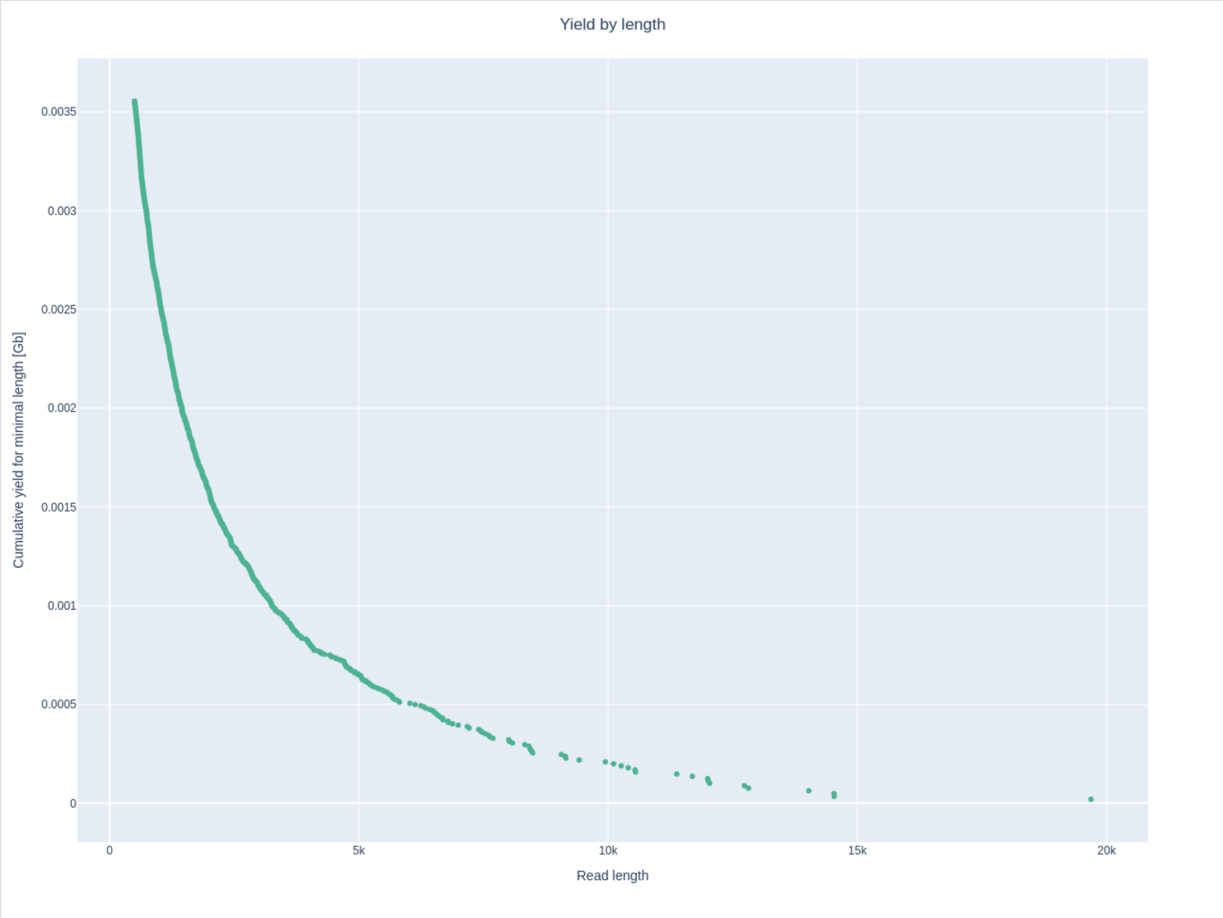


Post-Filtration QC

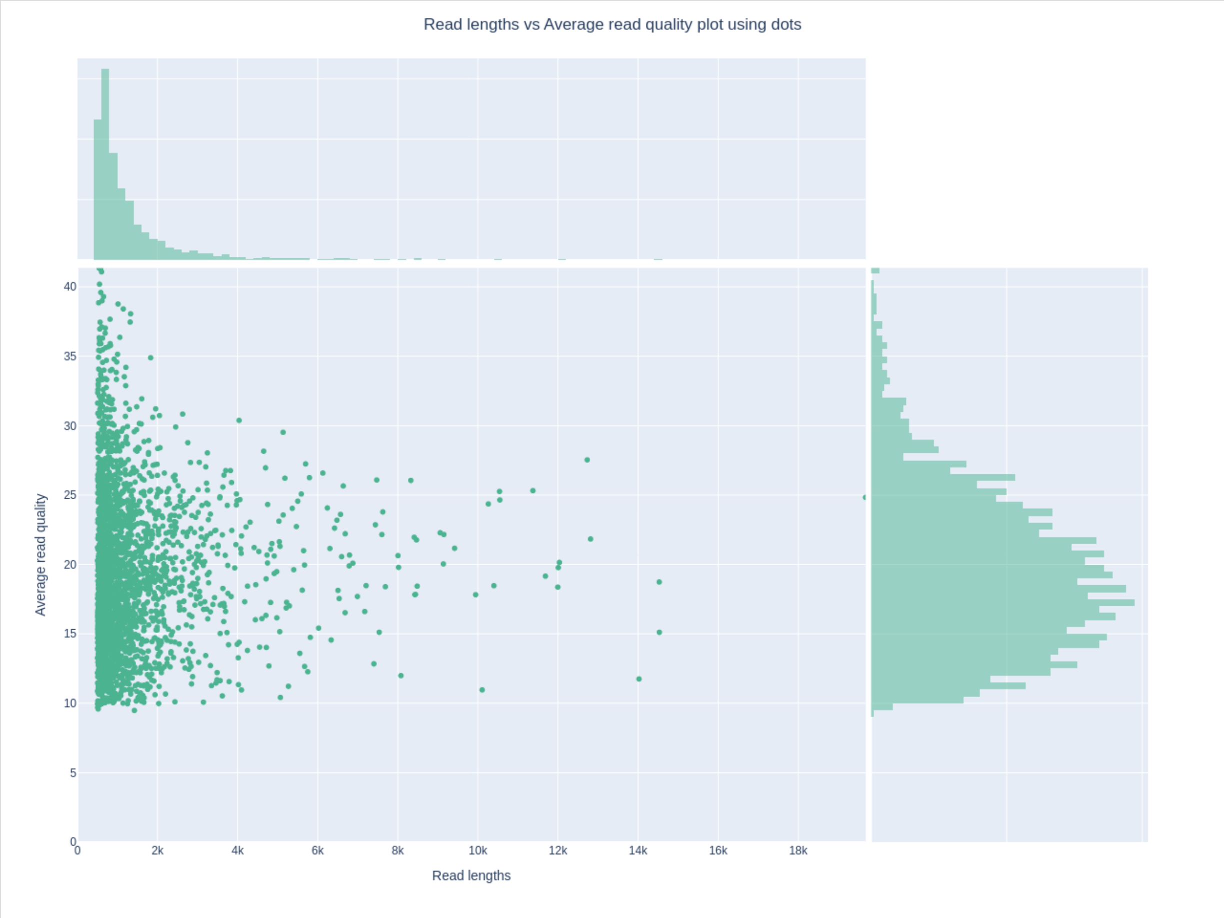
Statistics

```
General summary:
Mean read length:              1,385.2
Mean read quality:                16.7
Median read length:              878.0
Median read quality:              18.8
Number of reads:               2,566.0
Read length N50:               1,705.0
STDEV read length:             1,495.5
Total bases:               3,554,533.0
Number, percentage and megabases of reads above quality cutoffs
>Q10:   2557 (99.6%) 3.5Mb
>Q15:   1934 (75.4%) 2.8Mb
>Q20:   1087 (42.4%) 1.6Mb
>Q25:   409 (15.9%) 0.5Mb
>Q30:   114 (4.4%) 0.1Mb
Top 5 highest mean basecall quality scores and their read lengths
1:  41.4 (541)
2:  41.3 (581)
3:  41.1 (601)
4:  40.2 (550)
5:  39.6 (584)
Top 5 longest reads and their mean basecall quality score
1:  19684 (24.8)
2:  14534 (15.1)
3:  14530 (18.7)
4:  14026 (11.8)
5:  12816 (21.8)
```

Yield by Length



Length vs Quality



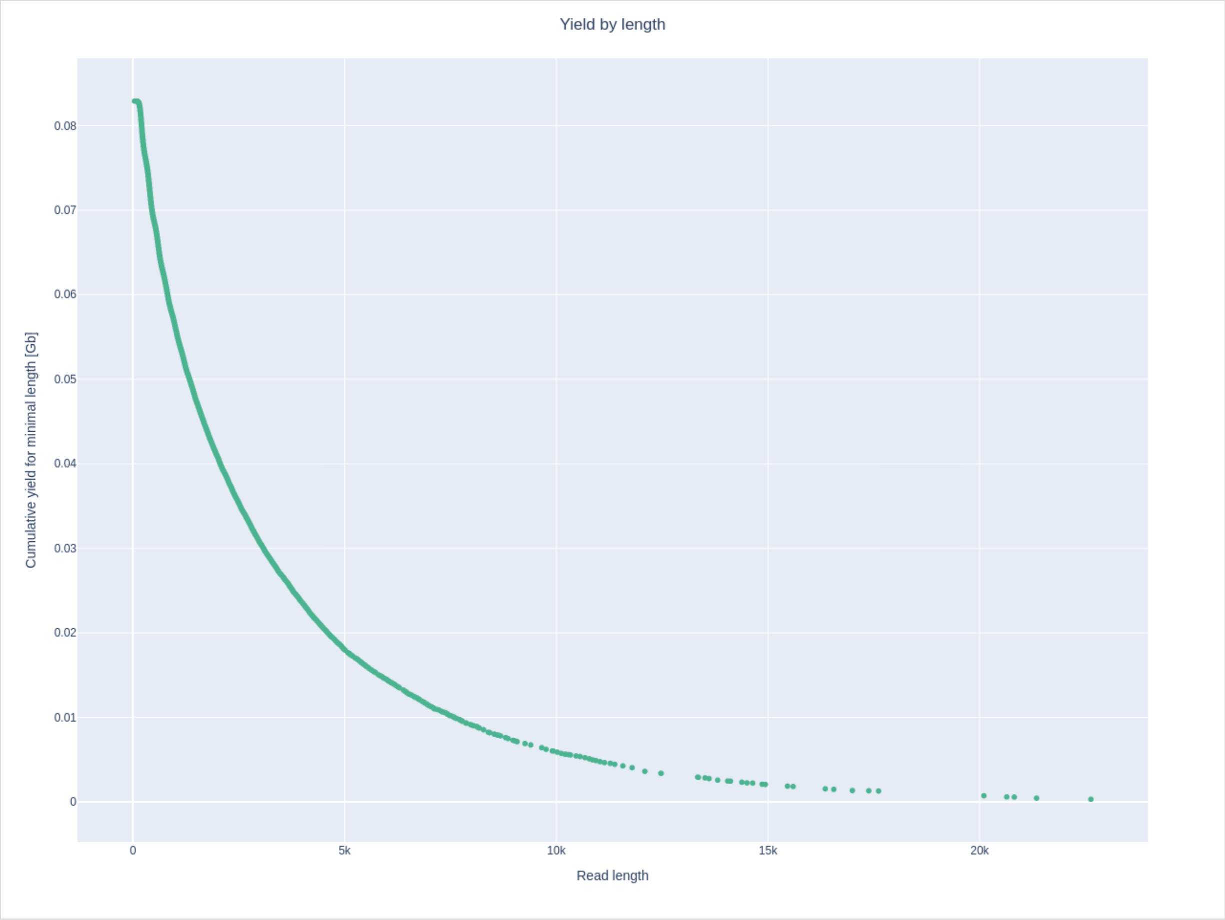
Sample: M1

Pre-Filtration QC

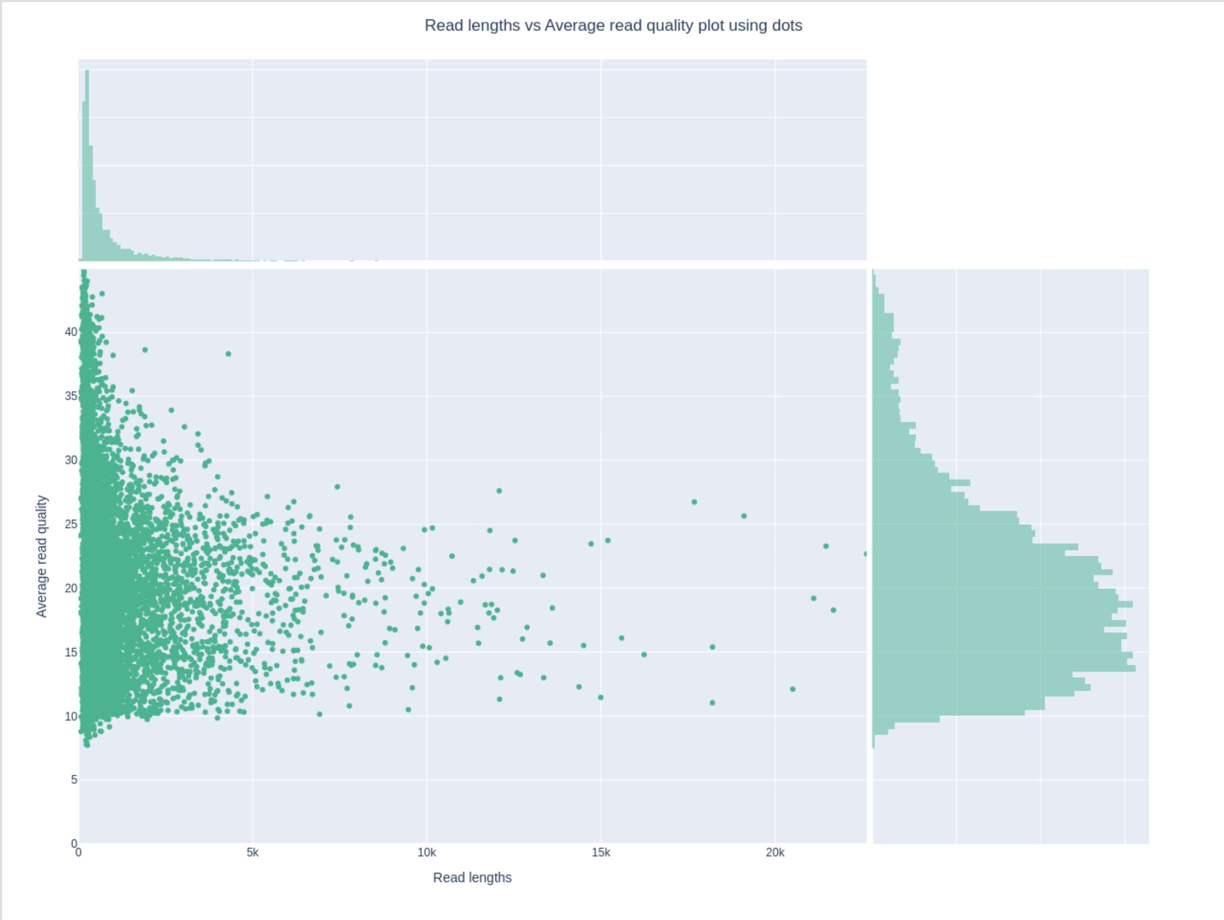
Statistics

```
General summary:
Mean read length:                 914.3
Mean read quality:                 16.5
Median read length:               409.0
Median read quality:               19.1
Number of reads:               90,663.0
Read length N50:                1,943.0
STDEV read length:              1,512.4
Total bases:               82,890,408.0
Number, percentage and megabases of reads above quality cutoffs
>Q10:	89580 (98.8%) 82.5Mb
>Q15:	67329 (74.3%) 62.9Mb
>Q20:	40454 (44.6%) 35.3Mb
>Q25:	18806 (20.7%) 11.7Mb
>Q30:	8351 (9.2%) 3.1Mb
Top 5 highest mean basecall quality scores and their read lengths
1:	45.6 (141)
2:	45.4 (228)
3:	45.1 (162)
4:	44.9 (153)
5:	44.7 (163)
Top 5 longest reads and their mean basecall quality score
1:	40666 (12.4)
2:	40563 (15.9)
3:	32913 (17.0)
4:	31255 (18.2)
5:	28296 (13.0)
```

Yield by Length



Length vs Quality

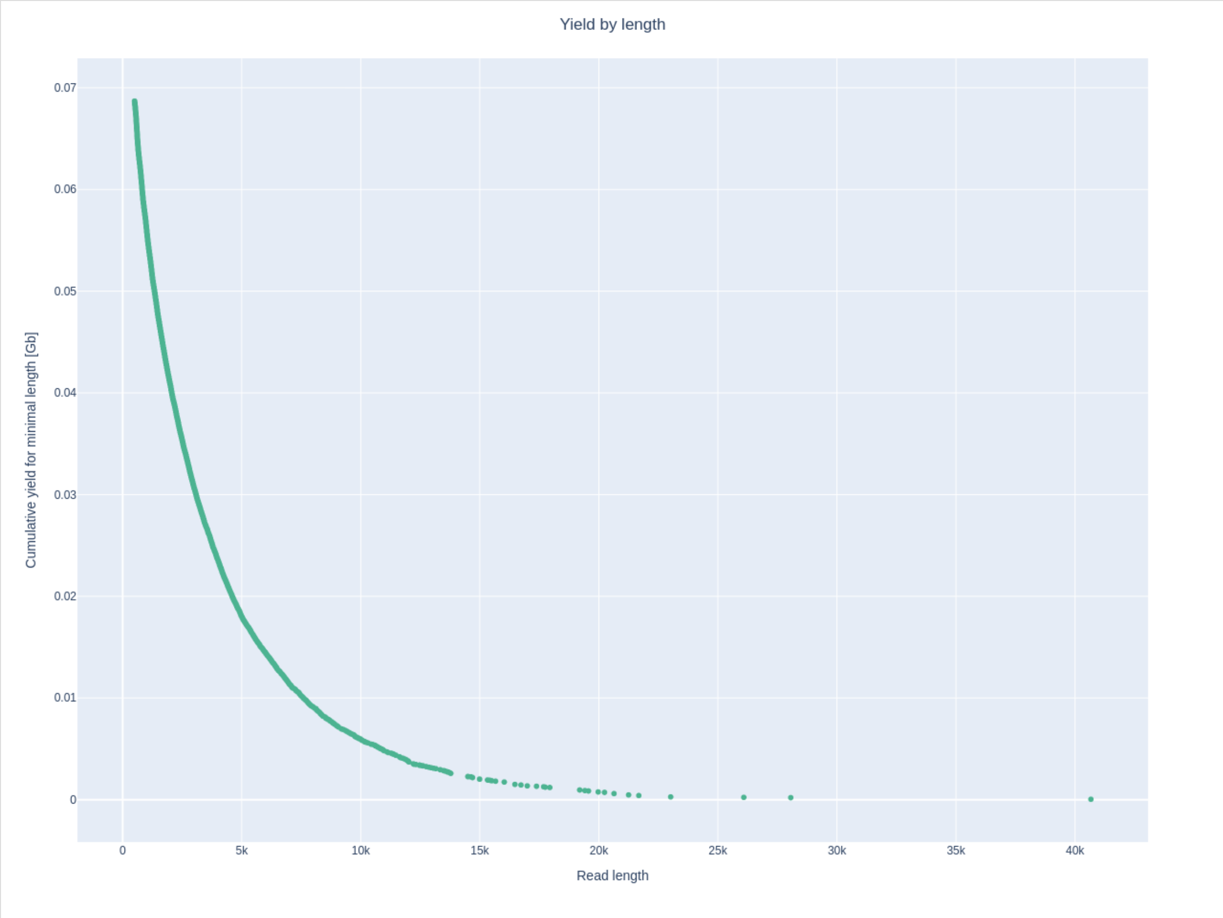


Post-Filtration QC

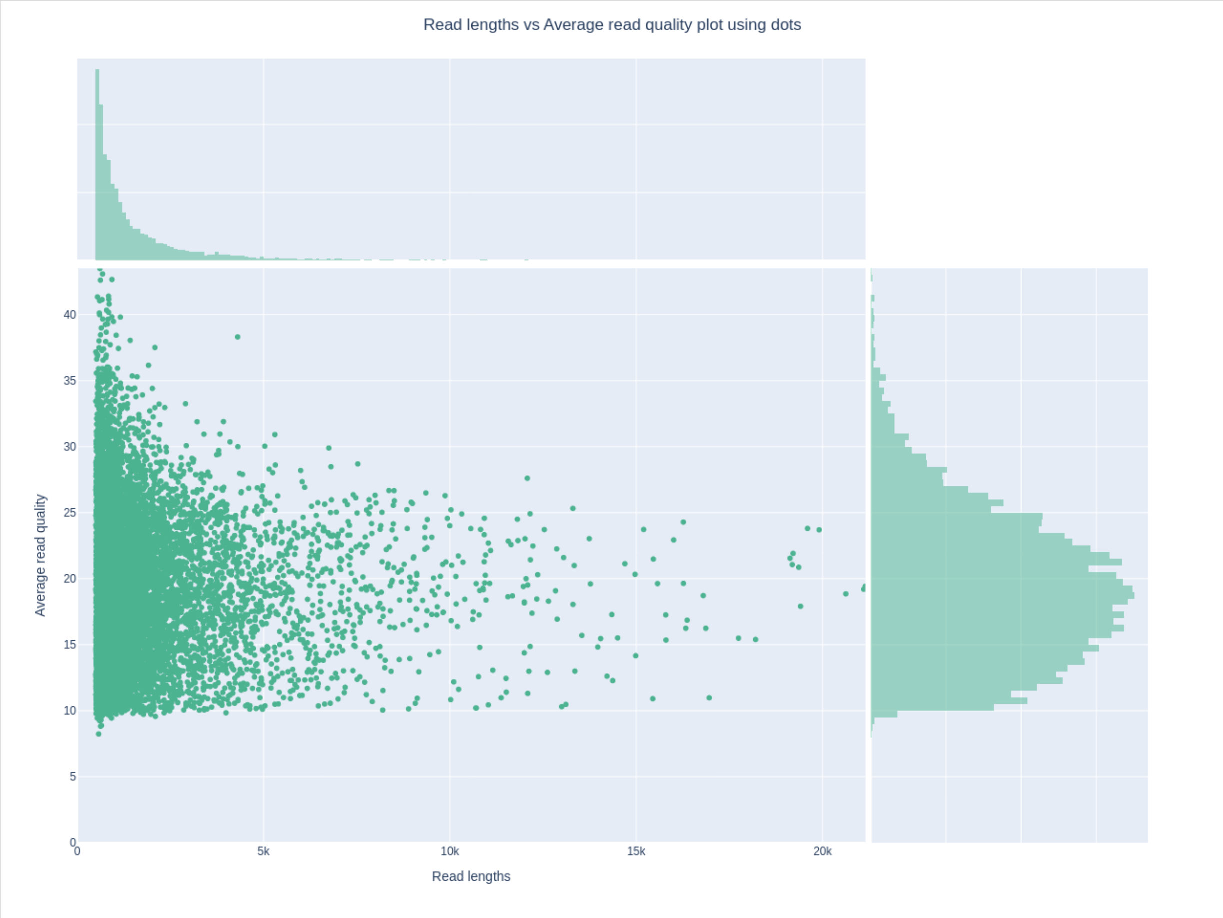
Statistics

```
General summary:
Mean read length:                1,784.3
Mean read quality:                  16.6
Median read length:              1,070.0
Median read quality:                18.9
Number of reads:                38,499.0
Read length N50:                 2,600.0
STDEV read length:               2,014.3
Total bases:                68,695,012.0
Number, percentage and megabases of reads above quality cutoffs
>Q10:   38320 (99.5%) 68.5Mb
>Q15:   29118 (75.6%) 52.6Mb
>Q20:   16474 (42.8%) 29.0Mb
>Q25:   5866 (15.2%) 8.4Mb
>Q30:   1486 (3.9%) 1.5Mb
Top 5 highest mean basecall quality scores and their read lengths
1:  43.5 (607)
2:  43.1 (680)
3:  43.1 (516)
4:  43.0 (680)
5:  43.0 (637)
Top 5 longest reads and their mean basecall quality score
1:  40666 (12.4)
2:  40563 (15.9)
3:  32913 (17.0)
4:  31255 (18.2)
5:  28296 (13.0)
```

Yield by Length



Length vs Quality



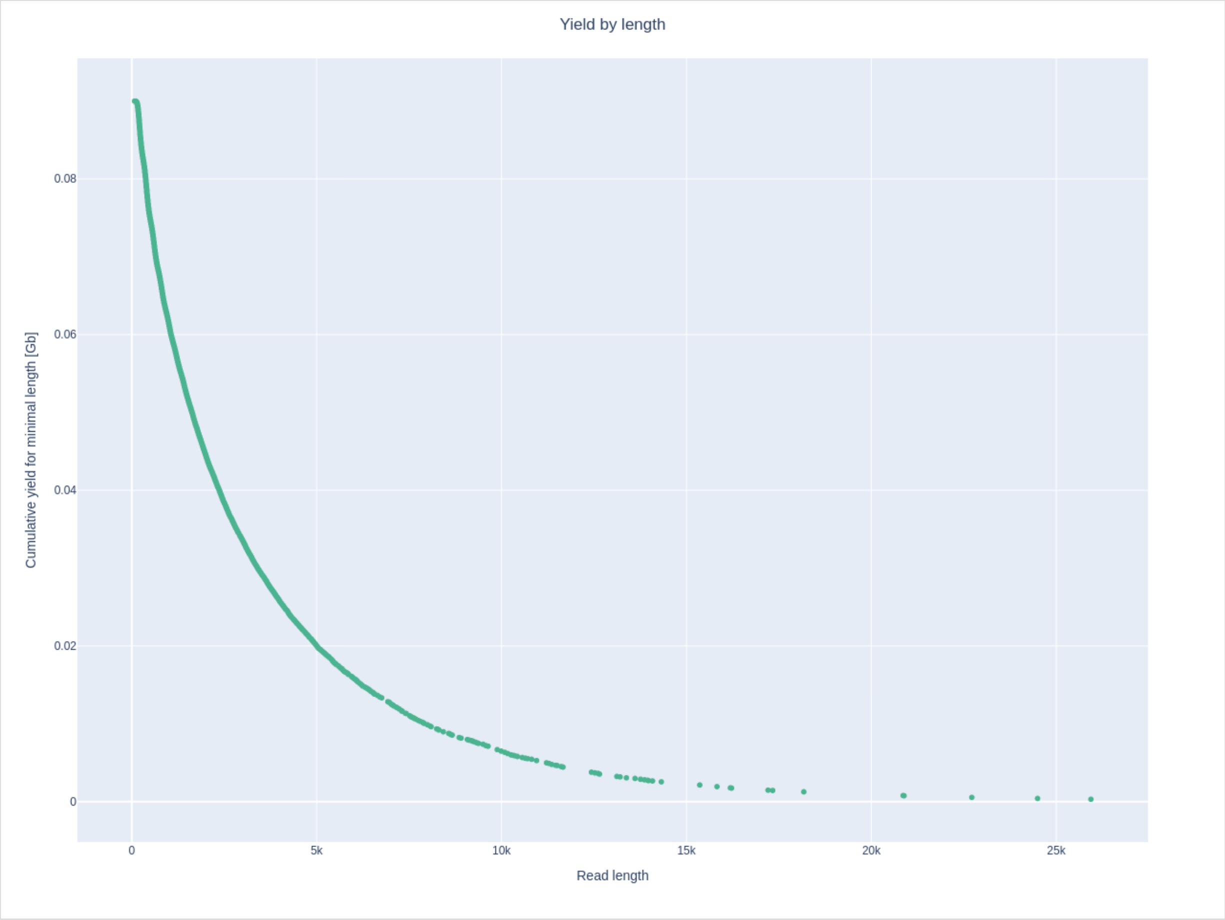
Sample: A1

Pre-Filtration QC

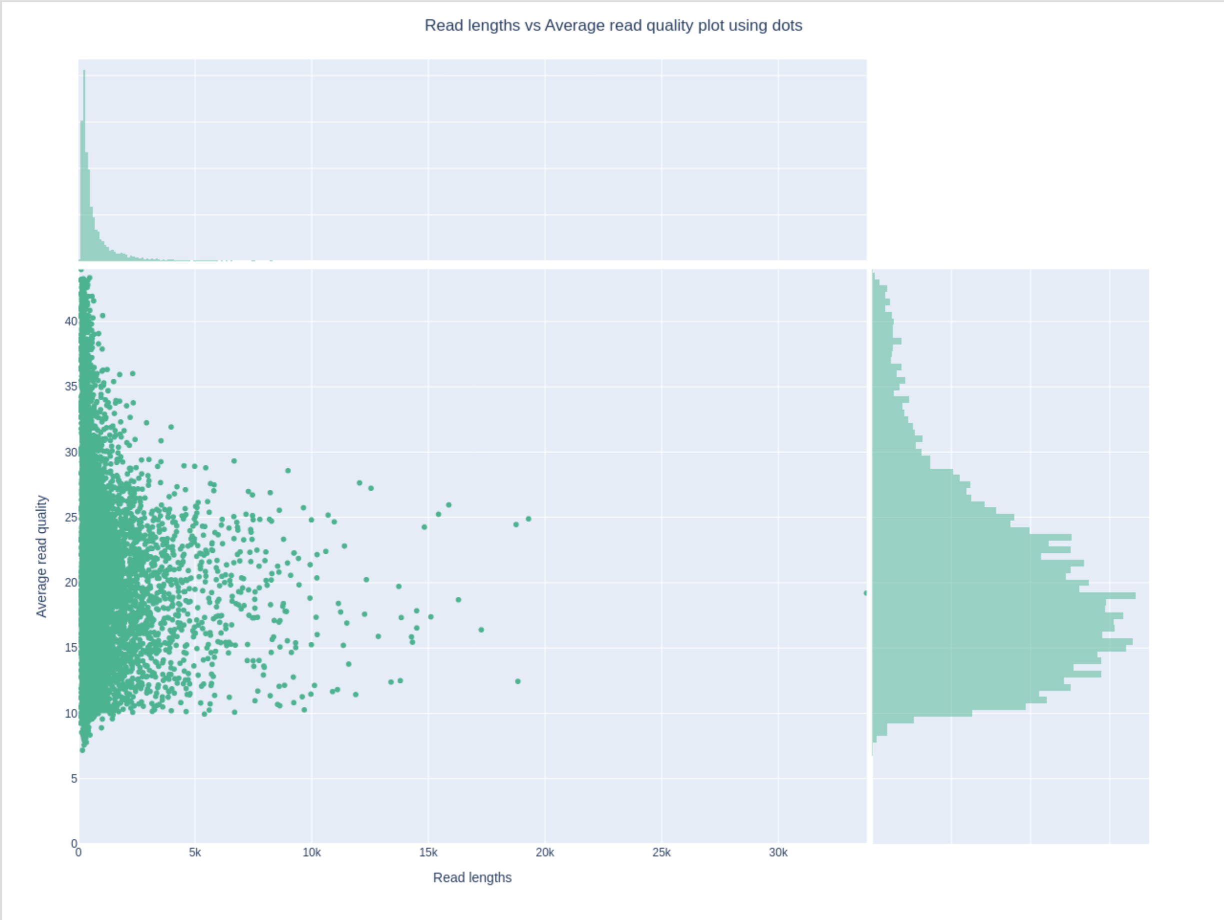
Statistics

```
General summary:
Mean read length:                 932.4
Mean read quality:                 16.4
Median read length:               417.0
Median read quality:               19.0
Number of reads:               96,525.0
Read length N50:                1,965.0
STDEV read length:              1,532.0
Total bases:               89,999,785.0
Number, percentage and megabases of reads above quality cutoffs
>Q10:   95330 (98.8%) 89.6Mb
>Q15:   70991 (73.5%) 69.1Mb
>Q20:   42499 (44.0%) 38.5Mb
>Q25:   19804 (20.5%) 12.5Mb
>Q30:   8940 (9.3%) 3.3Mb
Top 5 highest mean basecall quality scores and their read lengths
1:  44.8 (218)
2:  44.4 (150)
3:  44.4 (181)
4:  44.3 (232)
5:  44.3 (152)
Top 5 longest reads and their mean basecall quality score
1:  49492 (14.3)
2:  34666 (20.0)
3:  33788 (19.2)
4:  29264 (20.9)
5:  28957 (19.7)
```

Yield by Length



Length vs Quality

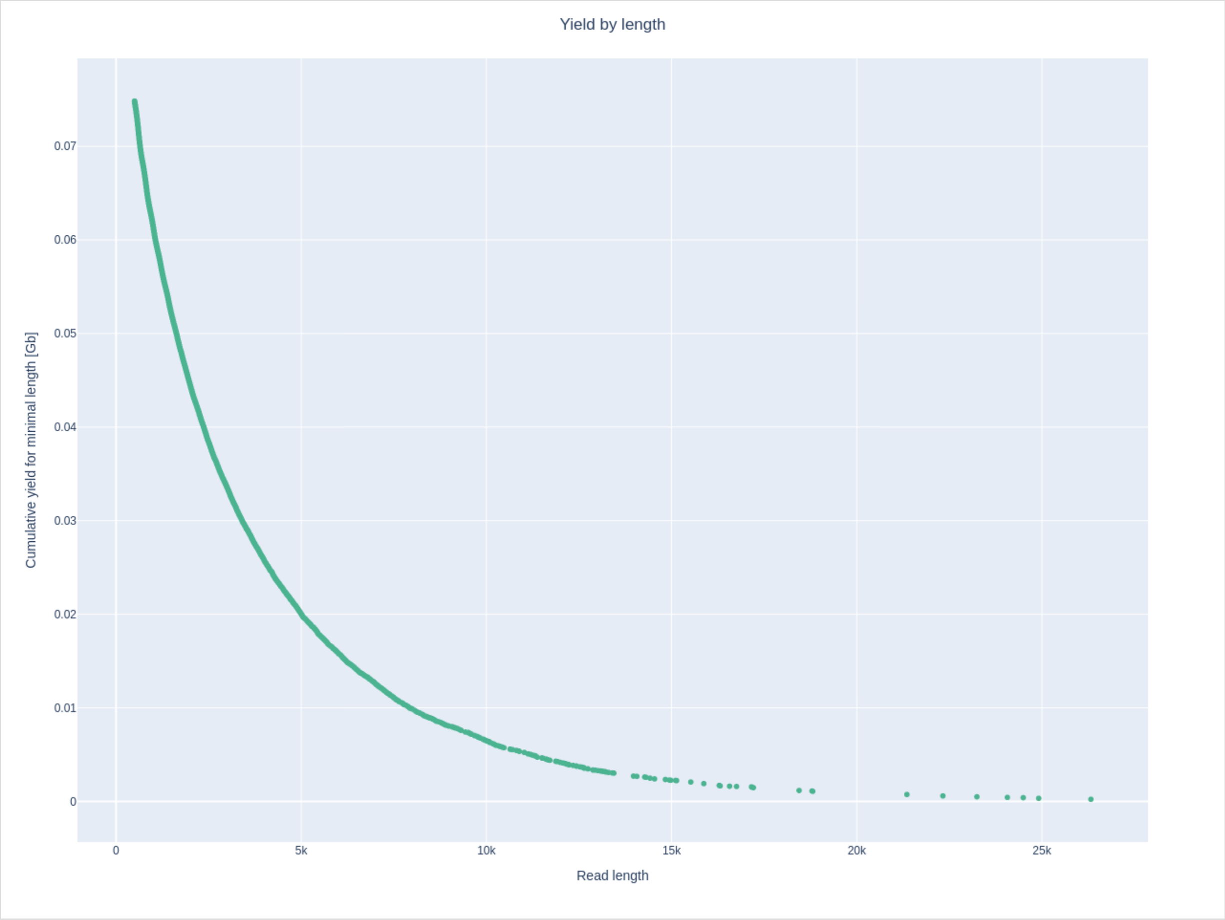


Post-Filtration QC

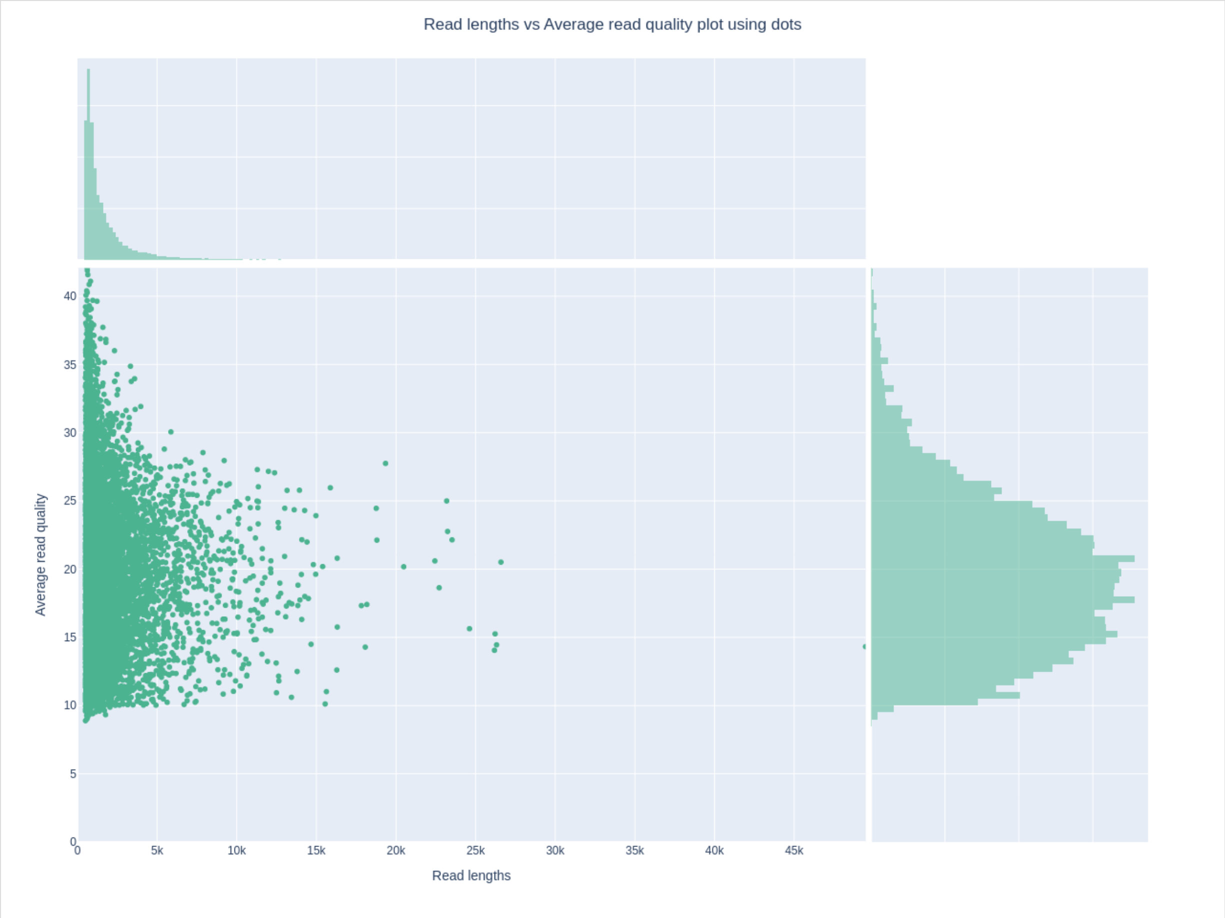
Statistics

```
General summary:
Mean read length:                1,799.1
Mean read quality:                  16.7
Median read length:              1,084.0
Median read quality:                18.9
Number of reads:                41,596.0
Read length N50:                 2,587.0
STDEV read length:               2,028.0
Total bases:                74,835,865.0
Number, percentage and megabases of reads above quality cutoffs
>Q10:   41436 (99.6%) 74.7Mb
>Q15:   31741 (76.3%) 58.4Mb
>Q20:   17787 (42.8%) 32.0Mb
>Q25:   6172 (14.8%) 9.0Mb
>Q30:   1598 (3.8%) 1.6Mb
Top 5 highest mean basecall quality scores and their read lengths
1:  43.9 (914)
2:  43.3 (518)
3:  43.0 (662)
4:  42.6 (1087)
5:  42.4 (558)
Top 5 longest reads and their mean basecall quality score
1:  49492 (14.3)
2:  34666 (20.0)
3:  33788 (19.2)
4:  29264 (20.9)
5:  28957 (19.7)
```

Yield by Length



Length vs Quality



Method:

QC pipeline steps:

```
1. Pre-QC Analysis: Raw reads analyzed with NanoPlot
2. Read Filtering: Reads filtered with FiltLong (min length: 500bp, min
mean Q: 12)
3. Post-QC Analysis: Filtered reads analyzed with NanoPlot
```

Analysis:

Explicacion tal vez del pq de los parametros

y algun comentario adicional dependiendo que sea bueno comentar por ejemplo

algun outlier en las muestras, los largos cortos, etc x.x.